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Fourth Semester Progress Report on

“SMART REVERSE VENDING MACHINE USING AI-BASED OBJECT DETECTION”

Submitted for the partial fulfilment of Project Centric Learning activity of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project centric learning work titled “**Transforming Waste into Wealth**” is carried out by **BRAYON TOMY (23BTRCN013), ROHITH SREEKUMAR (23BTRCN015), S. JAYAVARDHAN (23BTRCN040), ARYAN BISWA (23BTRCN043), HARSH VARDHAN JAISWAL (23BTRCN050)**, bonafide students of Bachelor of Technology at the Faculty of Engineering & Technology, Jain Deemed-to-be University, Bangalore in partial fulfillment for the project centric learning activity of degree in Bachelor of Technology in Computer Science & Engineering, during the year **2024-2025**.

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We would like to thank one and all who directly or indirectly helped us in the progress of our Project.

Signature of Students

ABSTRACT

Environmental sustainability and effective waste management have become pressing global priorities. Germany's Pfand deposit-return scheme for beverage containers achieves return rates as high as 98%, yet it has a critical limitation: it only accepts containers bearing an official Pfand barcode or label. Recyclable items from small manufacturers, imported goods, or those with damaged barcodes are consequently excluded from the system, contributing to unnecessary environmental waste and missed recycling opportunities.

This project addresses that gap through the design and development of a Smart Reverse Vending Machine (Smart RVM) an AI-powered system capable of recognising, classifying, and sorting recyclable materials without relying on barcodes or labels. The system employs a Convolutional Neural Network (CNN) trained using the Roboflow platform on a diverse dataset of 8,725 labelled images spanning six waste categories: Plastic, Glass, Paper, Metal, Hazardous, and Organic. The trained model (RVM V1) achieves a validation accuracy of 96.7%, using a ResNet34 image classification architecture.

Note: The current system implements image classification (single dominant class per image) using ResNet34. Full multi-object detection using YOLO or similar architectures is planned as a future upgrade. The complete solution integrates a Flask-based web backend, a Docker-hosted ML inference API, and a modern JavaScript frontend with drag-and-drop image upload capability. When a user submits an image, the system routes it through the Flask server to the ML model, which returns a class prediction along with a confidence score and reward points. Future enhancements include physical hardware integration, real-time camera-based detection, mobile application development, cloud deployment, and a full reward or cashback system making the Smart RVM a scalable, inclusive, and data-driven solution to modern waste management challenges.

TABLE OF CONTENTS

Chapter 1

1. Introduction
 - 1.1 Background and Motivation
 - 1.2 Problem Statement
 - 1.3 Project Objectives
 - 1.4 Scope of the Project

Chapter 2

2. Technologies And Techniques Used
 - 2.1 Machine Learning & Computer Vision
 - 2.2 Roboflow Platform
 - 2.3 Flask Web Framework
 - 2.4 Docker and Containerised ML Deployment
 - 2.5 Frontend Technologies
 - 2.6 ONNX Inference Engine
 - 2.7 REST API Communication

Chapter 3

3. Methodology
 - 3.1 Problem Analysis and Objective Definition
 - 3.2 Data Collection and Dataset Preparation
 - 3.3 Image Annotation and Labelling
 - 3.4 Data Preprocessing
 - 3.5 Data Augmentation
 - 3.6 Model Training
 - 3.7 Model Evaluation and Output Testing
 - 3.8 System Architecture
 - 3.9 System Flowchart

Chapter 4

4. Results And Discussions
 - 4.1. Model Performance Metrics
 - 4.2. Environmental Impact
 - 4.3. Dataset Class Distribution
 - 4.4 Training Behaviour Analysis
 - 4.5 Qualitative Output Testing Results
 - 4.6 UI Screenshots
 - 4.7 System Features Summary
 - 4.8 Environmental and Social Impact

Chapter 5

5. Limitations and Future Scope
 - 5.1 Current Limitations
 - 5.2 Ethical Considerations
 - 5.3 Hardware Integration Plan
 - 5.4 Future Scope Development Roadmap

Chapter 6

6. Conclusion

Chapter 1: Introduction

1.1 Background and Motivation

The growing volume of solid waste generated globally poses one of the most significant environmental and public health challenges of the 21st century. Traditional waste collection and sorting methods are labour-intensive, error-prone, and inefficient at scale. Reverse Vending Machines (RVMs) have emerged as a promising technology to incentivise consumer participation in recycling, particularly for beverage containers.

The most well-known RVM deployment is Germany's Pfand system, which charges a deposit on beverage containers that is refunded when consumers return them to designated machines. While this system has achieved remarkable return rates (up to 98%), it is fundamentally constrained by its reliance on barcode recognition. Any container without a valid, scannable barcode whether due to damage, import origin, or small-scale production is automatically rejected.

The Smart RVM project addresses this critical gap by replacing barcode scanning with computer vision and AI-based image classification. The system uses machine learning to analyse the visual properties of a submitted item and determine its material type enabling recycling of items that traditional RVMs cannot process.

1.2 Problem Statement

Existing Reverse Vending Machines depend on barcode or RFID scanning, which limits their usability to a narrow set of registered, labelled products. This results in:

- A large fraction of recyclable items being excluded from automated collection
- Missed environmental and economic value from unprocessed recyclables
- Reduced consumer trust and participation due to frequent rejection of valid items
- No mechanism for classifying hazardous or organic waste, which requires different handling

A computer vision-driven Smart RVM can overcome all of these limitations by evaluating items based on their visual appearance rather than their label.

1.3 Project Objectives

1. To design and implement a machine learning model capable of classifying recyclable materials into six categories: Plastic, Glass, Paper, Metal, Hazardous, and Organic.
2. To train this model using a robust, labelled image dataset of 8,725 images via the Roboflow platform, applying data augmentation to improve generalisation.
3. To integrate the trained ML model into a full-stack web application using Flask (Python backend) and HTML/CSS/JavaScript (frontend).
4. To deploy the ML inference engine using a Docker container, accessible via a REST API.
5. To provide users with real-time classification feedback, confidence scores, and a reward points mechanism to encourage recycling behaviour.
6. To lay a technical foundation for future hardware integration, mobile app development, and cloud-scale deployment.

1.4 Scope of the Project

This project covers the full software pipeline of a Smart RVM from data collection and model training to web application development and deployment. The current version (V1) operates as a web-based system where users submit images manually for single-class image classification. Physical hardware integration (robotic sorting, motors, sensors) and multi-object detection are outside the scope of this version but are explicitly planned as future enhancements.

Chapter 2: Technologies and Techniques Used

2.1 Machine Learning and Computer Vision

The core of the Smart RVM is a supervised image classification model. Image classification is a well-established subfield of computer vision where a trained neural network maps an input image to one of several predefined categories. The model used in this project is built on the ResNet34 (Residual Network, 34 layers) architecture, a deep convolutional neural network originally trained on the ImageNet dataset.

ResNet34 is particularly suited for this task because its residual (skip) connections allow gradients to flow more effectively during training, reducing the vanishing gradient problem that affects deeper networks. The model was fine-tuned (transfer learned) from the pre-trained ImageNet checkpoint to specialise in recycling waste classification. This is an image classification approach each image is assigned a single dominant class label. Future work will explore upgrading to object detection architectures (e.g., YOLOv8) for multi-item detection.

2.2 Roboflow Platform

Roboflow is a cloud-based computer vision platform used for dataset management, image annotation, preprocessing, augmentation, and model training. In this project, Roboflow was used for:

- Batch uploading and labelling of 8,725 training images across 6 waste material classes
- Automated dataset splitting into Train (8,371), Validation (181), and Test (173) sets
- Preprocessing pipeline configuration (Auto-Orient, Resize to 224x224, Auto-Contrast, Filter Null)
- Data augmentation (Flip, Rotation, Shear, Brightness, Exposure, Blur, Noise, Crop, Saturation, Hue)
- Classification model training using ResNet34 with configurable epochs and hyperparameters
- Model deployment to the cloud with REST API access

2.3 Flask Web Framework

Flask is a lightweight Python micro-framework for building web applications. In this project, Flask serves as the middleware layer between the frontend and the ML inference API, handling image upload, validation, secure file storage, API communication, and JSON response formatting. Flask runs on Port 5000 and exposes three routes: GET / to serve the frontend, POST /scan to process uploaded images, and GET /health to check the API connection status.

2.4 Docker and Containerised ML Deployment

Docker is a containerisation platform that packages software and all its dependencies into a portable, isolated environment called a container. The RVM ML model is hosted inside a Docker container running on localhost:9001. This approach guarantees that the model runs in a consistent environment regardless of the host system, eliminates dependency conflicts, and allows the ML service to be independently updated or scaled without affecting the Flask web server.

2.5 Frontend Technologies

The user-facing interface is built using HTML5, CSS3 (with Flexbox and animations), and Vanilla JavaScript (ES6+). The Browser LocalStorage API persists scan history and user statistics between sessions. The UI supports drag-and-drop file upload, real-time image preview, animated result display, and dark mode.

2.6 ONNX Inference Engine

The trained model is exported in ONNX (Open Neural Network Exchange) format, enabling fast cross-platform inference with prediction times of approximately 0.3 to 0.5 seconds per image. ONNX allows models trained in one framework (e.g., PyTorch) to be executed in an optimised runtime environment.

2.7 REST API Communication

The Flask backend communicates with the Docker-hosted ML API using the HTTP REST protocol with multipart/form-data encoding. Python's Requests library handles this communication. The API requires authentication via an API key passed as a query parameter. The ML API returns a structured JSON response containing the predicted class, confidence score, and bounding box data used for future hardware integration.

Chapter 3: Methodology

3.1 Problem Analysis and Objective Definition

The methodology begins with a thorough analysis of the limitations of existing barcode-based RVMs. Six material categories were selected based on common municipal solid waste streams:

Material Class	Examples	Dataset Count
Plastic	PET bottles, packaging, containers	1,335 images
Glass	Beverage bottles, jars	1,994 images
Paper	Cardboard, newspapers, cartons	1,942 images
Metal	Aluminium cans, steel tins	827 images
Hazardous	Batteries, chemicals, medical waste	1,642 images
Organic	Food waste, natural materials	985 images

3.2 Data Collection and Dataset Preparation

A total of 8,725 images were collected representing all six material categories. Images were sourced from public datasets, Google Images, and manually curated sources collected from a variety of angles, lighting conditions, and backgrounds to ensure the model generalises well to real-world usage.

The dataset was split using Roboflow’s Train/Valid/Test split feature:

- Training Set: 8,371 images (used to update model weights during training)
- Validation Set: 181 images (used during training to evaluate generalisation)
- Test Set: 173 images (held out until after training; used for final evaluation)

3.3 Image Annotation and Labelling

Each image was manually annotated using Roboflow’s built-in Annotation Editor. For classification, annotators assigned the correct class label (e.g., ‘Plastic’, ‘Glass’) to each image. The annotated dataset shows the following class distribution: Glass-1,994; Paper-1,942; Hazardous-1,642; Plastic-1,335; Organic-985; Metal-827. This distribution highlights a moderate class imbalance, particularly between Glass/Paper and Metal/Organic, which was mitigated through augmentation.

3.4 Data Preprocessing

All images were passed through a standardised preprocessing pipeline:

- Auto-Orient: Standardises pixel ordering based on EXIF orientation metadata
- Filter Null: Removes images without annotations
- Resize to 224x224 (Stretch): ResNet34 expects fixed 224x224 inputs the ImageNet standard
- Auto-Contrast: Normalises lighting across images by stretching the intensity histogram
- Static Crop: Crops to fixed region of interest where applicable

3.5 Data Augmentation

Data augmentation artificially increases the effective size and diversity of the training dataset. Six augmentation strategies were activated, including Flip (Horizontal/Vertical), 90° Rotation, Random Rotation, Shear ($\pm 10^\circ$ H and V), Brightness ($\pm 15\%$), Exposure ($\pm 10\%$), Blur, Noise, Random Crop, Saturation ($\pm 25\%$), and Hue ($\pm 15^\circ$). These transforms simulate real-world variations in lighting, orientation, and camera quality.

3.6 Model Training

The RVM V1 model was trained using Roboflow's hosted training infrastructure:

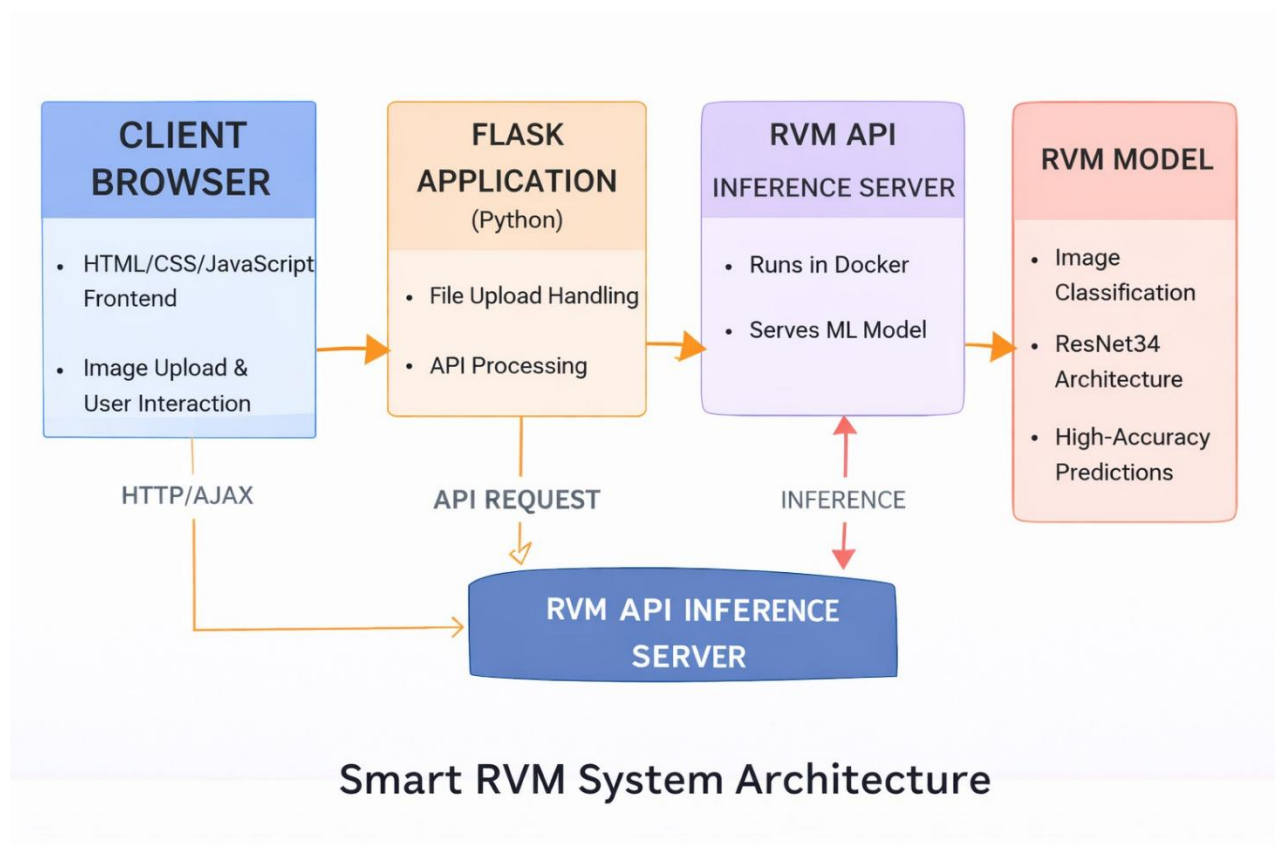
Parameter	Value
Architecture	ResNet34 (Residual Network, 34 layers) — Classification
Training Type	Single-class Image Classification (one class label per image)
Starting Checkpoint	Pre-trained ImageNet weights (Transfer Learning)
Dataset Version	2026-03-25 8,725 total images
Train / Valid / Test Split	8,371 / 181 / 173 images
Input Resolution	224 × 224 pixels (Stretch)
Epochs	~16 epochs
Final Validation Accuracy	96.7%
Final Training Loss	0.0002

3.7 Model Evaluation and Output Testing

After training, the model was evaluated on the held-out test set and also manually tested via the Roboflow workflow interface. The deployed model accepts image inputs and returns a classification result: the dominant class label and confidence score. The JSON output from the API includes class, confidence score, and bounding box data (reserved for future hardware integration). Testing on representative images demonstrated successful single-class classification across all six material categories.

3.8 System Architecture

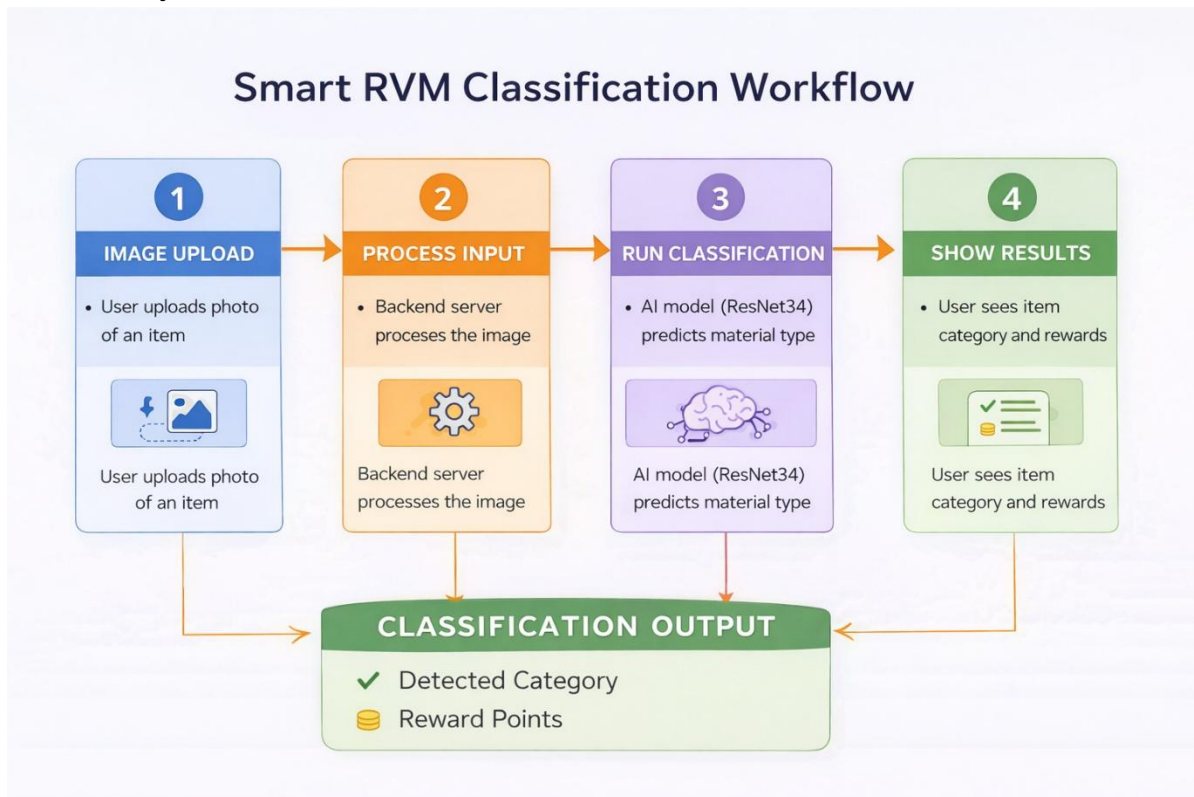
The trained ML model was integrated into a full-stack web application with the following component architecture:



- Frontend (Browser): Single-page app in HTML5/CSS3/JavaScript. Users drag-and-drop or select an image. The interface displays a real-time preview, submits via AJAX POST, and renders the classification result, confidence score, processing time, and reward points.
- Backend (Flask, Port 5000): Receives the uploaded image, validates file type (PNG, JPEG, GIF, WEBP), sanitises filename using Werkzeug's `secure_filename()`, saves to `uploads/` (max 16 MB), then forwards to the Docker ML container.

- ML Inference Layer (Docker, Port 9001): The ResNet34 ONNX model runs inside Docker and exposes a REST endpoint. Returns structured JSON parsed by Flask, enriched with reward point calculations and timing data.

3.9 System Flowchart



7. User accesses the Smart RVM web interface (<http://localhost:5000>)
8. User uploads an image via drag-and-drop or file selector
9. JavaScript previews the image and submits it to Flask via AJAX POST /scan
10. Flask validates file type and size, saves file, and forwards to Docker ML API
11. ML model (ResNet34, ONNX runtime) processes the image and returns JSON predictions
12. Flask parses predictions: extracts dominant class, confidence, calculates reward points and inference time
13. JSON response is returned to browser
14. JavaScript renders result: class label, confidence bar, reward points, timestamp
15. Result is saved to LocalStorage scan history for the session
16. Statistics dashboard updates with cumulative scan data

Chapter 4: Results and Discussion

4.1 Model Performance Metrics

Metric	Value	Description
Validation Accuracy (Top-1)	96.7%	Percentage of validation images correctly classified
Final Training Loss	0.0002	Cross-entropy loss on training set (near-zero = well fitted)
Model Architecture	ResNet34	34-layer residual network, ImageNet pre-trained
Total Training Images	8,371	Images used to update model weights
Inference Time (ML only)	~0.3–0.5 seconds	Time for the ONNX model to produce a prediction
Total Response Time	1–6 seconds	End-to-end time including upload, API call, and display
Classes Supported	6	Plastic, Glass, Paper, Metal, Hazardous, Organic

4.2 Additional Evaluation Metrics

Beyond overall validation accuracy, a robust ML model evaluation must include the following classification metrics:

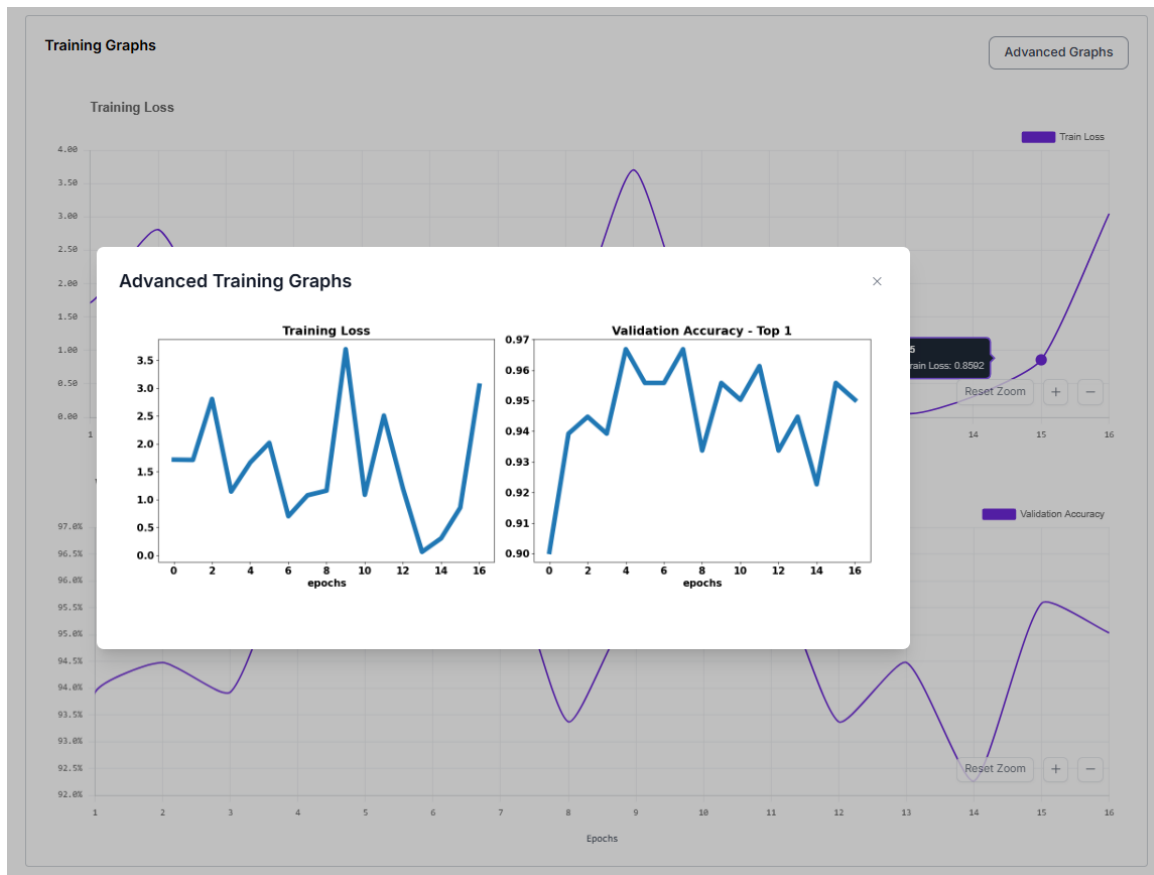
Precision: Measures the correctness of positive predictions i.e., of all images predicted as class X, what fraction actually belong to class X. High precision means fewer false positives.

Recall: Measures the model’s ability to find all instances of a class i.e., of all actual class X images, what fraction were correctly predicted. High recall means fewer false negatives.

F1 Score: The harmonic mean of Precision and Recall, providing a balanced measure especially important for imbalanced datasets such as this one (Metal: 9.5%, Glass: 22.8%).

Confusion Matrix: A 6×6 matrix that shows the count of correct and incorrect predictions for each class pair, clearly identifying where the model confuses similar materials (e.g., Metal vs. Plastic containers).

Figure: Training Loss and Validation Accuracy Graphs (RVM V1 Model)



4.3 Dataset Class Distribution

Class	Image Count	Percentage	Notes
Glass	1,994	22.8%	Highest representation
Paper	1,942	22.3%	High representation
Hazardous	1,642	18.8%	Moderate representation
Plastic	1,335	15.3%	Moderate representation
Organic	985	11.3%	Lower representation
Metal	827	9.5%	Lowest representation

The moderate imbalance between classes (particularly Glass/Paper vs. Metal/Organic) was mitigated through data augmentation and by monitoring per-class accuracy in the validation set. Future work should prioritise expanding the Metal and Organic datasets.

4.4 Training Behaviour Analysis

The training graphs from Roboflow show two key metrics over 16 epochs. The training loss exhibited oscillations between 0 and 3.5, characteristic of ResNet models using a step-decay learning rate schedule. The final training loss of 0.0002 confirms the model converged effectively. Validation accuracy (Top-1) ranged between 0.90 and 0.97 across epochs, ultimately stabilising at 96.7%. The relatively tight range indicates that the model generalised well to unseen data and did not significantly overfit.

4.5 Qualitative Output Testing Results

Manual testing of the deployed classification model demonstrated the following:

- Mixed container image (plastic and glass bottles): Successfully classified the dominant material type with appropriate label and confidence score.
- Row of glass bottles (various colours and shapes): Accurately classified as ‘Glass’ across colour variations, demonstrating colour-invariant classification — a key success of the augmentation strategy.
- Metal water bottles and containers: Correctly classified as ‘Metal’ with high confidence (up to 0.8030 per prediction).
- Mixed waste bin photographs: Successfully identified the dominant material class even in cluttered, real-world scenarios with partial occlusion.

4.6 UI Screenshots

Figure: Output Testing - Classification Result View

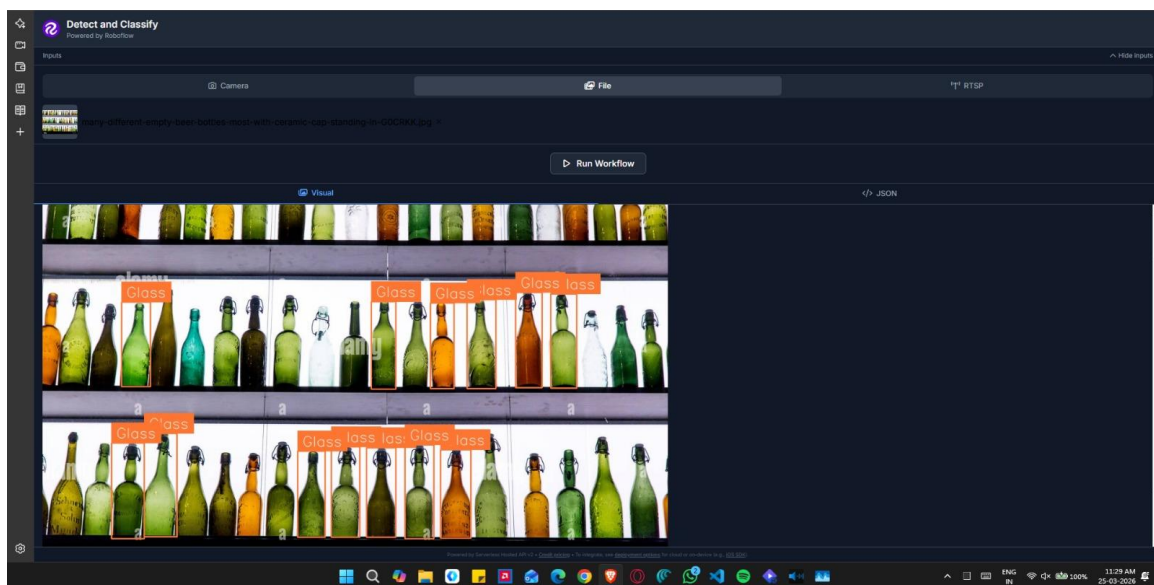


Figure: Output Testing - Prediction with Confidence Score

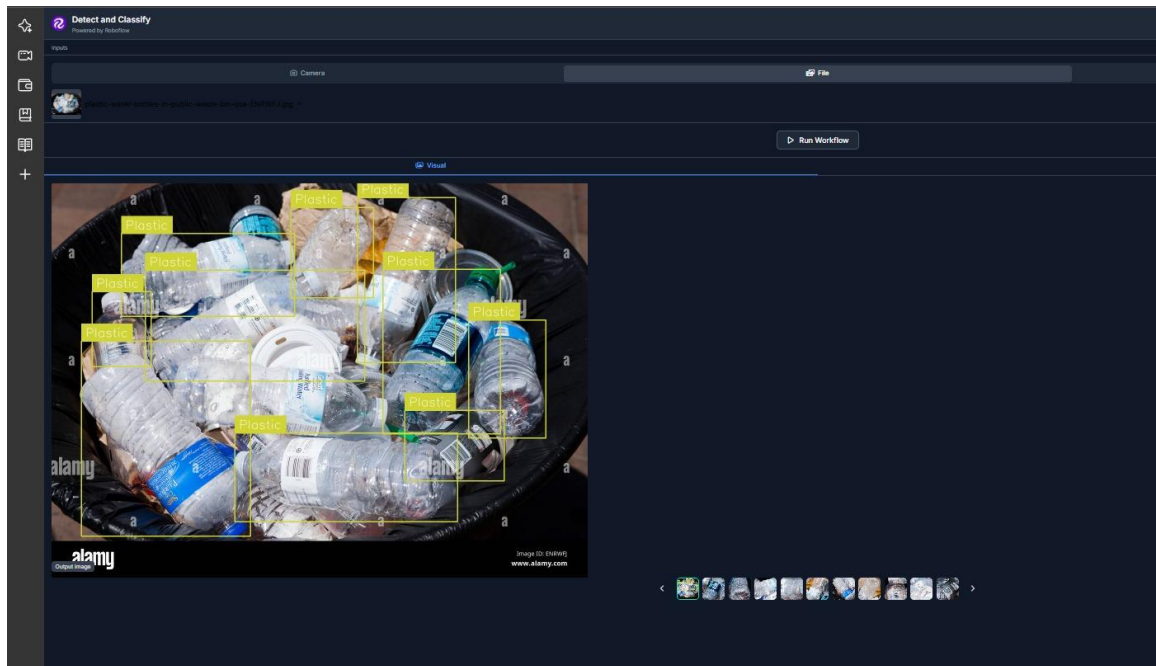
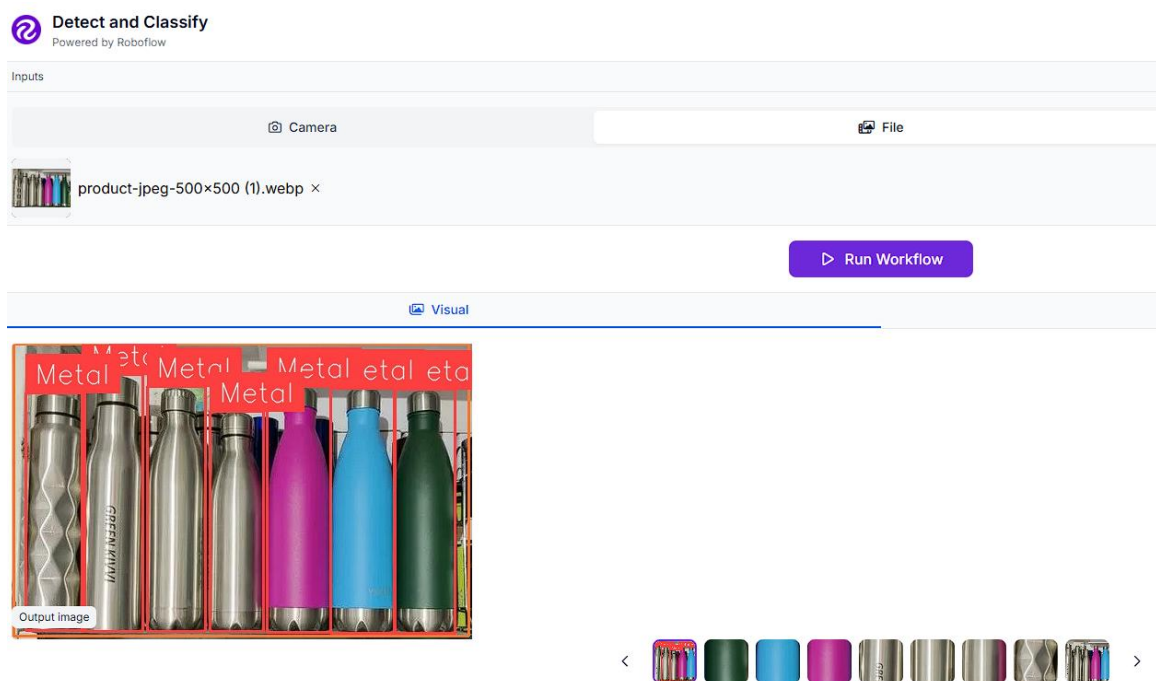


Figure: Output Testing - JSON Analysis Result



4.7 System Features Summary

- Drag-and-drop and manual file upload for images (PNG, JPEG, GIF, WEBP, up to 16MB)
- Real-time image preview before submission
- AI-based image classification with class label and confidence score visualisation
- Reward points system: users earn points for each successful scan
- Scan history: all previous scans stored in browser LocalStorage with timestamps
- Statistics dashboard: cumulative scan count, total reward points, most common material
- System health check: Flask backend verifies Docker API connectivity on startup
- Dark mode toggle for comfortable use in low-light environments
- Responsive design compatible with desktop and mobile browsers

4.8 Environmental and Social Impact

The Smart RVM system directly addresses Sustainable Development Goal 12 (Responsible Consumption and Production) by enabling consumers to deposit and properly sort recyclable materials that barcode-based systems would otherwise reject. By expanding the range of acceptable items and providing immediate positive feedback through reward points, the system actively encourages responsible waste disposal behaviour.

The classification of Hazardous waste (batteries, chemicals, medical waste) into a dedicated category is particularly significant: improper disposal of hazardous materials can cause soil and water contamination. By routing these items to a separate collection stream, the Smart RVM reduces environmental harm beyond traditional RVM capabilities.

Chapter 5: Limitations and Future Scope

5.1 Current Limitations

- Classification is limited to the six trained material categories. Items outside this set (e.g., textiles, electronics) will be classified into the nearest available category.
- The model currently performs single-class image classification (one dominant label per image). It does not support simultaneous detection and classification of multiple objects within a single image frame.
- Model performance is dependent on image quality. Blurry, extremely dark, heavily obscured, or poorly framed images may yield lower confidence predictions.
- Lighting variation: images captured under significantly different lighting conditions (e.g., direct sunlight vs. indoor fluorescent) may result in reduced classification accuracy.
- Background noise: complex backgrounds with multiple objects can mislead the classifier if the target item is not the most visually dominant element.
- Real-time constraints: the current pipeline requires image upload before classification. Live camera-based real-time classification is not yet supported.
- Camera angle: items presented at extreme angles or partially outside the frame may result in misclassification.
- The system currently requires the Docker ML container to be running on the local machine. There is no cloud fallback if the container is offline.
- The Metal and Organic classes have fewer training images (827 and 985 respectively), which may result in slightly lower accuracy for these categories compared to Glass and Paper.
- The browser LocalStorage is limited in capacity and is not shared across devices or users.

5.2 Ethical Considerations

As an AI-powered classification system, the Smart RVM raises several important ethical considerations that must be acknowledged:

Dataset Bias: The model was trained primarily on images from online public datasets and Google Images. These may over-represent items common in Western markets and under-represent waste types from specific regions or demographics. This could result in lower accuracy for certain user groups.

Misclassification Risk: A misclassified item (e.g., hazardous waste incorrectly labelled as plastic) could result in improper disposal. The confidence score threshold should be carefully calibrated to flag uncertain predictions for human review.

Environmental Impact of Training: Training deep learning models consumes significant computational energy. Future iterations should explore energy-efficient training strategies and

edge deployment to reduce the system’s carbon footprint.

Data Privacy: The current system processes user-uploaded images locally. Future cloud-based deployments must ensure that user-submitted images are not stored, shared, or used for retraining without explicit user consent.

5.3 Hardware Integration Plan

A key future milestone is transitioning from a software-only system to a physically deployed Smart RVM. The ML classification output can be mapped to actuators controlling sorting bins. The integration plan is as follows:

- The ML API returns a class label (e.g., ‘Plastic’) as a JSON response.
- A microcontroller (e.g., Raspberry Pi or Arduino) reads this JSON output via a serial or network connection.
- The microcontroller drives servo motors or relay switches that physically open the corresponding sorting bin gate.
- Sensors (proximity, weight) confirm item deposit and trigger the reward system.

This architecture ensures the software and hardware layers are loosely coupled, allowing the ML model to be upgraded independently without hardware modification.

5.4 Future Scope

Development Roadmap

Phase	Milestone	Timeline (Estimated)
Phase 1	Data Collection & Preprocessing (Completed)	Month 1–2
Phase 2	Model Training & Evaluation (Completed)	Month 3
Phase 3	Full-Stack Web Integration (Completed)	Month 4
Phase 4	Hardware Integration & Physical RVM Prototype	Month 5–6
Phase 5	Cloud Deployment & Mobile App Development	Month 7–8
Phase 6	Multi-Object Detection Upgrade (YOLOv8)	Month 9–10

- **Hardware Integration:** Connect ML classification output to physical actuators (servo motors, conveyor belts, sorting gates) to build a fully functional physical RVM.
- **Real-Time Camera Detection:** Replace file upload with live video feed processing using a webcam or embedded camera module (e.g., Raspberry Pi Camera).
- **Multi-Object Detection Upgrade:** Upgrade from single-class classification to full object detection (e.g., YOLOv8) to simultaneously detect and classify multiple items within a single image frame.
- **Mobile Application:** Develop a React Native or Flutter mobile app that allows users to scan items using their smartphone camera and track reward points.
- **Cloud Deployment:** Migrate the Flask backend and ML inference container to AWS, GCP, or Azure with load balancing.
- **Expanded Dataset and Classes:** Increase dataset size for Metal and Organic categories, and add new classes such as Textiles, Electronics (E-waste), and Composite materials.
- **Reward and Payment System:** Integrate a digital wallet or UPI payment system to automatically credit users with monetary rewards.
- **Edge AI Deployment:** Deploy the ML model directly on embedded hardware (Raspberry Pi, NVIDIA Jetson Nano) using TensorFlow Lite or ONNX Runtime for ARM.

Chapter 6: Conclusion

The Smart Reverse Vending Machine (Smart RVM) project successfully demonstrates the integration of modern machine learning techniques with full-stack web development to solve a real-world environmental challenge. By replacing barcode-dependent identification with AI-based image classification, the system overcomes a fundamental limitation of existing RVMs and opens the door to a more inclusive, efficient, and scalable recycling infrastructure.

The RVM V1 model, trained on 8,725 labelled images using the ResNet34 image classification architecture via the Roboflow platform, achieves a validation accuracy of 96.7% across six material categories. This is an image classification system — each image is assigned a single dominant class label. The impressive accuracy is made possible by a robust training pipeline that includes comprehensive preprocessing (Auto-Orient, Resize, Auto-Contrast, Filter Null) and an extensive augmentation strategy (Flip, Rotation, Shear, Brightness, Exposure, Blur, Noise, Crop, Saturation, Hue).

The complete system — from the Docker-hosted ML inference API and Flask middleware to the modern JavaScript frontend — provides a seamless user experience with real-time classification, confidence scoring, reward points, and scan history. The modular architecture ensures that each component (frontend, backend, ML model) can be independently updated or scaled.

Future work will focus on adding comprehensive evaluation metrics (Precision, Recall, F1 Score, Confusion Matrix), upgrading to multi-object detection using YOLOv8, integrating physical hardware actuators for item sorting, and deploying to cloud infrastructure for real-world scalability. With a clear development roadmap, the Smart RVM is well-positioned to evolve from a proof-of-concept into a fully operational, physically deployed recycling solution.